2026 **HIGHER SCHOOL CERTIFICATE EXAMINATION**

Mathematics Advanced

General Instructions

- Reading time – 10 minutes
- Working time – 3 hours
- Write using black pen
- Calculators approved by NESA may be used
- A reference sheet is provided at the back of this paper
- For questions in Section II, show relevant mathematical reasoning and/or calculations
- Write your Centre Number and Student Number at the top of this page and on the Question 11 Writing Booklet attached

Total marks: 100

Section I – 10 marks (pages 2–8)

- Attempt Questions 1–10
- Allow about 15 minutes for this section

Section II – 90 marks (pages 9–40)

- Attempt Questions 11–31
- Allow about 2 hours and 45 minutes for this section

Section I

10 marks
Attempt Questions 1–10
Allow about 15 minutes for this section

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1 What are the square roots of $-3 + 4i$?

A. $1 - 2i$ and $-1 + 2i$

B. $1 + 2i$ and $-1 - 2i$

C. $3 + i$ and $-3 - i$

D. $-3 + i$ and $3 - i$

Section II

90 marks
Attempt Questions 11–31
Allow about 2 hours and 45 minutes for this section

For questions in Section II, your responses should include relevant mathematical reasoning and/or calculations.

Question 11 (5 marks)

(a) The complex numbers w and z are given by $w = 3e^{i\pi/2}$ and $z = e^{i\pi/3}$. **2**

Find the modulus and argument of wz^2.

(b) The complex number z is given by $5 + 2i$.

Find, in Cartesian form:

(i) z^2 **1**

(ii) z^{-1} **2**

Question 12 (3 marks)

Given the function $f(x) = x^2 + 3x - 4$, find $f''(x)$. **3**

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